

To What Extent are Users Able to Recognise Personality Traits from a Chatbot?

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Abstract—Chatbots have become used in many applications in recent years to supply more interactive user experiences to websites, customer service and virtual assistants. Research has increased focus on this topic to create more realistic chatbots to talk to and to expand their use cases. In recent years, there has been a big expansion into conversational Artificial Intelligence (AI) to simulate human-like conversations. This paper investigates whether users can recognise a chatbot's personality traits according to the HEXACO scale. The chatbot was made using a chatbot powered by the AI DeepSeek. Participants then talked to the chatbot and answered a HEXACO observer form with their perception of the chatbot's personality. This study found that participants could not recognise any of the HEXACO traits with accuracy, with the average value for the participants ranking the traits as middling value. These findings suggest the idea that users cannot recognise traits from a purely text based conversation and that other avenues like facial expressions or tone of voice should be experimented with. The results from this study imply that the focus of chatbot development should be on improving efficiency and training rather than focusing on personality.

Index Terms—Chatbot, Personality, AI, HEXACO, DeepSeek

I. INTRODUCTION

IN recent years, the use of Artificial Intelligence (AI) and Machine Learning (ML) has increased dramatically. AI-based chatbots have become integrated into various services such as virtual assistants, customer support and mental health support. These systems aim to improve user experience by generating quick and human-like responses to queries.

A chatbot's personality can affect user engagement and the level of trust. However, from existing research, it is unclear if users can accurately tell all 6 personality traits within a chatbot, as defined by the HEXACO personality model. Previous studies have relied on outdated 5 factor models while focusing on one or two traits, leaving a gap in research on how AI personalities are interpreted across all six traits. To address this research gap, a chatbot powered by DeepSeek AI with a fixed personality was developed. Each trait has a different emphasis in the generated responses. Participants then talked to and filled out a HEXACO observer form[1] to assess their perception of the chatbot's personality.

Finding if users can recognise personality traits has a strong impact on chatbot and AI development and design. If a solely text-based interaction cannot convey personality then other methods such as facial expressions or text to speech with voice tones could be used to better increase the realism of a chatbot. A lack of recognition would also signify that development time is being better spent elsewhere outside of creating a unique personality for each chatbot, such as response time optimisation.

II. LITERATURE REVIEW

This chatbot allows for back-and-forth conversation, with the chatbot being able to answer questions and ask questions. Compared to traditional chatbots that just ask questions and store user answers.[2] This is useful for many reasons since it allows for a more natural and human conversation to take place, with each member of the conversation being able to ask and answer different details and questions about themselves. The traditional chatbot model only asks questions, and the user fills the role of only giving answers, which does not have a natural flow as it cannot branch off from the set questions the chatbot has to follow to traditionally solve the issue that the user has.

This paper used popular conversational AI models to test them using the Myers-Briggs test to find out their personality type.[3] This test categorises people into personality types from a questionnaire. From the findings, they found that these models had consistent personality types. This could be due to the data that is used to train the model, as the language and sentences used can affect how the questions will be answered. Conversation AI and chatbots can be trained using all kinds of different data. Usually the data will need to be accurate to human conversations, so things such as movie scripts and subtitles are perfect since they were written to be spoken and to be believable.[4]

Chatbots can be trained on very specific data to obtain an outcome that is wanted and needed to fulfil a specific task. This can be used to create bots that help teach course-specific material to students.[5] This works reliably alongside traditional routes of teaching and the material being given to the chatbot to be a study companion that helps students find the answers. On the other hand, companies are researching and developing conversational AIs that are being used to create engaging game characters.[6][7] This shows the versatility that chatbots and conversational AI have and how helpful they can be for learning, working and entertainment.

By utilising Large Language Models and conversational AI, data can be stored so that answers that are within the correct context can be given to users.[8] Conversational intelligence could also be applied to achieve an active conversation and to have an awareness of the topic that is being discussed.[9] This works well in tandem with teaching these models from movie scripts as it provides lots of different lines and contexts to teach the model how to communicate like a human.[10]

This paper is based on a competition where human-like chatbots are made and judged. It describes the process of making the bots using ChatScript[11], which is an open-source

language. They used this language to create a chatbot with a personality to win this competition, discussing the methods, issues and more that will be relevant in this paper. This paper is similar to what is being researched, but data is not shown within the paper. It primarily discusses the creation of this chatbot and what issues they ran into while running it at an event, where data was not collected from users other than the chat logs.[12]

DeepSeek is an open-source Large Language Model(LLM).[13] This was developed as a rival and alternative to other generative AI that is available. The main creation from this company is the DeepSeek-r1, the reasoning model. It was created to find training limitations, testing what can be done and what the limits are for LLM models.[14] Within the research, it was found that DeepSeek outperformed other models that had more parameters. These findings make this model unique as it is the most advanced, easily accessible AI model that can be used within research. As an open source software it is important to consider for research and the availability that it gives.

Ollama is an open-source software that allows a simple way to load and run LLMs locally.[15] It creates a virtual server that is responsible for the management and downloading of different models. This software was created with the intent of giving greater privacy and control than alternatives that use online services. LangChain is an open-source software that helps with the development of LLM-powered applications.[16] This allows for models to access data that is outside of their training, creating applications that have a wider use case. It also allows for prompt engineering to be implanted into the model along with user requests to ensure that the model behaves the way that is needed.

Prompt Engineering is where inputs are designed and optimised to ensure that responses are generated in the desired way. Prompt engineering improves LLMs performance when completing a specific and custom task.[17] By utilising this, developers can create more realistic responses that remain relevant to the context of the solution. This reduces the number of responses that are labelled as incorrect as the model has the necessary context to answer inputs within the situation that is required.

Creating attachment and relationship with AI chatbots such as Replika.[18] Replika is an app that gives users access to AI-powered chatbots. Users can talk to these at any time, getting a personalised interaction that is aimed to replicate human interaction. Social chatbots can be used for mental health as users can develop an attachment to these bots, receiving emotional support and encouragement from them.[19] This study found that individuals can develop attachment to a chatbot if they gain positive interaction from it. Some users within this study would talk to the Replika chatbot daily and would experience distress when separated from it.

The creation of a function within AI-based NPCs that can be extended to chatbots is the existence of MemoryRepository. This functions as a short-term and long-term memory for conversational models. The majority of models have bad recollection and quickly forget what was said. This can make them frustrating to talk to as they will repeat responses, and

the user will get into a loop of getting the same reply. This paper proposes and creates a way to fix this issue by using human-like recollection and summarisation. This creates a more human and natural conversation for users.[20] When AI, chatbots and other virtual agents make use of dynamic and shifting personalities, it becomes difficult for the user to keep in mind a consistent identity. When adding visual context, it ties together the perception for the user by connecting it to a visual component like a 3D model or drawing.[21]

In this paper, it was proven that human-computer connection is improved by keeping the same information and identity across devices.[22] Having identifiable factors like brand names helps users connect the same agent on different devices, keeping a strong continuity of identity for the bot within the user. When the identity and information did not migrate to different devices, the user's perception of this agent dropped. This research allows insight into the ability of a person to recognise certain traits from a chatbot and conversational AI, but what is not fully tested within this paper is where this human recognition lies.

This paper looked into the effect of a person talking to a publicly available conversational AI. This research led to people feeling connected to this agent, having feelings of attachment and forming a friendship.[23] This allows an insight into how well humans can connect to a technology like this, as the participants within this study knew it was not a real person that they were talking to, but at the end of this study, they still showed intent to keep talking to this AI. A problem that this paper found was social stigma, as people were concerned that the wider public would react negatively to finding out that someone used an AI model as a conversation partner.

The creation of a chatbot with a personality is possible.[24] The technique used within this paper utilised a "personality component" to create acceptable responses. Within this research, the normal creation process for a chatbot is used with added features like a decision-making process to generate answers and a personality database that included basic characteristics for the chatbot. These characteristics are things like family connections, food preferences, and the emotional component that is used. Within this paper, the strength of the personality is tested by comparing the responses and data generated by the chatbot to a previously made version. The newly developed chatbot was then run with and without the emotional component against one participant. Having a personality component that takes longer to generate a better, more accurate reply is better since users do not mind slightly extended response times for a more accurate response.[25]

This paper focuses on human-robot interactions within a healthcare context. This paper reviewed the usage of healthcare robots and whether the usage of a personality made them more effective and trustworthy within this field.[26] This paper takes a strong focus into the healthcare usage of this technology, but still provides relevant insight towards the research field as a whole. As a review of many papers across the field, it shows that many papers focus on the impact of an extroverted personality on an introverted personality. It also discusses the lack of research within the interactions of human-robot

personalities and how each is affected within a medical field. As this paper is not based within a healthcare sector, not all points fully apply, but it still provides great context into where this technology can be used to further research and develop solutions.

Chatbots can also be used for things such as psychological and mental health care.[27] In this paper, chatbots that can help deal with emotions using trained methods from therapists are discussed. As pointed out in the paper, many people are not willing to seek help for these issues, so having a chatbot help a user in these situations separates from the possibly embarrassing human-human connection. If a chatbot could be given an easily identifiable personality, these bots could become successful, as people would feel comfort from this.

A chatbot with a personality has been made before, where the personality shifts according to how the user rates the conversation.[28] It starts off with a base personality and periodically asks the user to rate how they feel the conversation is going. This rating then determines how the chatbot evolves, changing the language and the way it communicates with the user. Within this paper, the only personality types reviewed are Extroversion and Agreeableness from the OCEAN model[29]. This study only targeted the usage of two of the five models, so in their future work section, they mention adding more personality patterns to account for more of the personality range.

The OCEAN model is now considered outdated and has been replaced by the HEXACO model.[30] This model uses 6 points of measurement, so it is considered more accurate. The HEXACO model consists of: Honesty-Humility, Emotionality, Extroversion, Agreeableness, Conscientiousness and Openness. Having an extra degree of measure over the OCEAN model makes HEXACO more precise in its measurements. The Five Factor OCEAN model has papers documenting it dating back to 1993. Since HEXACO is a newer model, there is proof that it considers a wider range of things, and it aims to fix any shortcomings that OCEAN has.

The usage of chatbots for customer service has been debated for some time, if a chatbot decreases the value of customer service or if it in fact brings in more customers.[31] In this paper, the psychology of this was reviewed and tested to see what customers thought of chatbots. Overall, they found that chatbots were good for negative interactions with customers like a failure in service. If a chatbot communicated that to the user, then the majority of people within this study were found to be more understanding. The main issue they found with using chatbots within customer service is that many participants stated that they prefer to talk to other people and that the emotionless nature of a chatbot could harm customer relations as there is no sincerity behind what is being said. By giving emotions to a chatbot, the overall user experience is improved.[32] This is due to the bot having a more humanoid way of interacting, which increases the level of trust that a user has, this overall improves quality as the user believes in more what was said and feels reassured that it is a reliable service.

Through the usage of this trust that can be created through a humanised product, long-term business and repeat customers

and usage can be created.[33] Since the bot is seen as more human, people associate more human traits with what the bot can do. Personality is a strong factor in the humanisation of a chatbot, but it is not the only option. By changing the database to focus on different language, the bot can seem like a different person.

III. RESEARCH QUESTION AND HYPOTHESES

A. Research Question

The literature review shows that lots of research and thought has been put into all manner of subjects within chatbots and conversational AI. In previous research, it is shown that users have more positive attitudes towards chatbots that have more humanised conversation styles. It is also shown that more and more people are researching into the field of extended use of this type of technology. With research into this technology being used within games, customer service, healthcare industries and more, it creates a plethora of research opportunities to help this technology evolve. After reviewing all the literature, this paper looked into the question *How accurately do participants identify HEXACO personality traits from a personality-focused chatbot?* This is due to there being a gap in literature, with not many papers that researched this topic in detail.

B. Hypotheses

The independent variable within this study was the participants, as I will be collecting data from many people. The dependent variable is the personality of the chatbot. Using a t-test[34], the distance between the opinions of the participants and the value of the personality can be measured. The hypotheses could be generated from previous works that proves that participants can recognise agreeableness within a chatbot. The hypotheses of this experiment are listed within table III, with one for each personality trait being measured with a corresponding null hypotheses.

IV. COMPUTING ARTEFACT

A. Description

This paper's research goal involved getting participants to engage with a chatbot within a predefined context. The chatbot they talked to was programmed with certain personality traits as listed by the HEXACO personality model. The HEXACO personality model is a psychological model that focuses on behaviour and personality through the use of 6 identifiers. Within this study, participants were asked to have an introductory conversation with the chatbot, asking them questions about what they think of different ideas and concepts. The participants asked these questions to gain a measurement on how it would reply and what personality they pick up from it. During the experiment, the participants will be given a scenario to act out with the bot to ensure that the messages they get will be useful in forming their opinions. This scenario repeated the experiment information they were given within the information sheet, and they were tasked with asking questions to get to know the chatbot well enough to assume things about it. Once the participant asked enough questions

Symbol	Null	Symbol	Alternative
H01	Agreeableness is recognised in the chatbot	H1	Agreeableness is not recognised in the chatbot
H02	Conscientiousness is recognised in the chatbot	H2	Conscientiousness is not recognised in the chatbot
H03	Emotionality is recognised in the chatbot	H3	Emotionality is not recognised in the chatbot
H04	Extraversion is recognised in the chatbot	H4	Extraversion is not recognised in the chatbot
H05	Honestly-Humility is recognised in the chatbot	H5	Honestly-Humility is not recognised in the chatbot
H06	Openness to Experience is recognised in the chatbot	H6	Openness to Experience is not recognised in the chatbot
H07	Personality on average is recognised in the chatbot	H7	Personality on average is not recognised in the chatbot

TABLE I

A TABLE OF THE NULL AND CONNECTED ALTERNATE HYPOTHESES

to where they were confident to answer the questionnaire and make assumptions they moved on to filling out a questionnaire. The questionnaire used the sixty-question HEXACO observer form. Once this data had been collected and translated to numerical values, these could then be graphed and statistically tested.¹

B. Methodology

This section talks about the developmental and experimental process that is used within this study to see the extent of recognition users have from chatbot personalities. This will go through each step of the creation within this study, detailing different areas and software used. It will then go on to mention how testing and the participant process was handled.

The project was made by utilising python scripts, libraries, Ollama and Deepseek's reasoning model. This bot will utilise the HEXACO personality model to simulate a personality into the responses and for the participants to review the chatbot that they spoke to. This ensures accuracy, as utilising different personality measures would create unreliable data as traits are defined with different parameters per personality model. The hardware used within this project consisted of a desktop computer. This computer had 32GB of RAM at 3000MHz, AMD Ryzen 9 5900x 12-core CPU, and a NVIDIA 3070ti with 8GB of VRAM. The computer also made use of SSD storage for quick access. The steps that were used within this project are:

- 1) Installing Ollama and Deepseek
- 2) Integration with Python
- 3) Prompt Engineering
- 4) Unit and AB testing
- 5) Collecting participant data
- 6) Review, analyse and discuss data

1) *Installing Ollama and DeepSeek:* Ollama is an open-source software created to make Large Language Models (LLMs) easily used and modified by the general public. This software was vital to this study as it allowed for the research to use state of the art technology easily and without having to pay expensive fees. Using Ollama allows for downloadable LLMs to be run locally on available hardware, creating an easily integrated LLM. This does come with the downside of being limited by the available hardware. The LLM model has to be downloaded onto the storage and, when ran, uses the computer memory and graphics card to logically think

to provide a response. Using Ollama, one of the newest LLMs that can be used is called DeepSeek. This was recently released to the time of this study and was intended as a competitor to ChatGPT which is considered the state of the art. Once Ollama was downloaded, the different levels of models could be loaded onto the local hardware. A model with 8 billion parameters was used within this research. This gave the best cost of performance to quick response time with the hardware that was used throughout this study.

2) *Python Integration:* Once the model and Ollama had been selected it could be connected through a python script to allow for better user interaction and customisation. Using an open-source software called LangChain, the Python script could better interact with the model. LangChain is designed to make prompt engineering more efficient by connecting the prompt and the user question. This was a vital tool for creating this model to research from, as it led to the creation of a chatbot to be much simpler than expected since the prompt and user input could be inserted into the locally ran LLM. Without LangChain, the LLM model would have to be retrained or fine-tuned using datasets that would have to be collected and the training tested. This retraining method would have been outside the scope of this study. Once LangChain and Ollama had been integrated into the python script, allowing the activation of the model, information and messages to be passed into the LLM development moved onto creating an application that was more user friendly. Tkinter was used to create a graphical user interface (GUI) allowing participants to have an easier and more familiar experience talking to the chatbot. This involved creating an input text box where the user could type, a send button and a conversation log that shows the messages that they send and what the chatbot replies with. With all these pieces combined, it created an easy-to-use application that linked the locally run LLM model using Ollama through LangChain to connect to the Python frontend, allowing it to be passed to a user interface. It was decided to create a cleaning function for the model output. This is due to DeepSeek being a reasoning model that outputs its thought process. This meant that for each message sent, multiple lines were outputted describing how the model used reason and patterns to get to the selected answer before outputting the answer. The cleaning function takes the thought process that is between two think tags and removes it before the model's output is shown to the user. This function uses the built-in Python re library. One line of code is used to search for the think tags. Once a match is found, the resulting think tags and everything between them are removed to give a clean output to the user.

¹<https://github.falmouth.ac.uk/GA-Undergrad-Student-Work-24-25/Comp302-2203238.git>

3) *Prompt Engineering*: Prompt Engineering was a useful tool within this experiment. As mentioned earlier, it reduced the need for fine-tuning within a study with a short time frame. This involves giving the AI LLM a specific task to complete before the user interacts with it. It is instructed to remain following that task at all times while it responds to the user inputs. Giving the AI chatbot a prompt to initialise it with ensures that it remains consistent within the answers that it gives, ensuring it can complete the task that it has been created to do. This the case of this study the prompt given to the chatbot instructs it on how to act, with what personality traits and that it should reply to the users inputs in a human like manner. Without the use of prompt engineering a custom dataset would have to be created and used to ensure the bot acted and replied a certain way.

4) *Testing*: After development had been completed, the created chatbot was tested. This is talked about in further detail within section VI of this paper. Unit tests were used to check that the core functions of the chatbot were working. This included external libraries and that the response of the chatbot was working. AB testing also took place. This involved altering the prompt fed into the chatbot, example questions were used to collect answers from chatbots with two different prompts. The one that adhered to the personality desired the most was then selected. This continued until a satisfactory prompt was found that gave the best result compared to all others completed.

5) *Data Collection*: Participants were gathered to talk to the chatbot and fill out a questionnaire in response to the conversation that they had. Participants were given access to the information sheet, the chatbot and a text document giving them a scenario to perform and example questions to ask. Since this study involved seeing if users could tell the personality from a chatbot they were asked to imitate the scenario of meeting someone for the first time. The questions they were provided were simple examples that they could use while also coming up with their own. For example, one of the questions was to ask the chatbot if they enjoyed large groups or smaller groups. From these example questions and the scenario given, participants could effectively interact in a meaningful way that would allow them to take part in the study. Once the participants felt that they had had a satisfactory conversation, they were allowed to move on to the questionnaire. This data collection form was the official HEXACO personality reporter form; the sixty-question HEXACO observer form was used as this allows for proper data collection while not requiring too many questions that the participant is sat answering for a prolonged period. This asked the participant to make assumptions based on the conversation by asking the participant to rate how likely and if the chatbot would enjoy different scenarios and actions. The participant ranked these on a Likert scale, going from strongly disagree to strongly agree as per the HEXACO standards. If the participant could tell the personality from the conversation, they would answer the questions correctly to get an answer that supports the hypotheses.

6) *Data Analysis*: After the data had been collected within the Microsoft form that hosted the HEXACO form, the par-

ticipants' answers were able to be changed into HEXACO scores. For this, the HEXACO mark scheme was used to go through and change the participants' Likert scale scores from being based on a strongly disagree to strongly agree scale to a numerical scale going from one to five. This had to be done this way due to the nature that the HEXACO form works; some of the questions have the scores inverted due to the language of the questions. After these scores for each facet of each trait had been calculated, they could be added up to give each personality trait an average score from all participants. These values could then be mapped against the hypothesised values that were created when the chatbot was given levels of each personality. Using R and RStudio, t-tests could then be ran using the collected values to find out the statistical significance of these values as well as graph them against the hypothesised value to show the difference between the hypothesised values and the values that were found from the research.

V. SYSTEM LIFECYCLE

A. Planning & Analysis

This research paper fills a gap in the literature, as pointed out by the literature review earlier on. Literature that was found that performed similar research to this ran the study on very few participants, causing a difficulty in finding good statistical analysis within them as there is not a large enough data collected. Other papers that looked into this research focused on the effects a chatbot with a personality could have within the customer service industry, specifically on the amount of negative customer reviews that could be reduced. This paper left a gap in the research since it was proven that there was a positive impact made within that study but not if the personality could be recognised. Users may have recognised the personality or the humanisation of the chatbot enough to improve the usage and affect the outcome. The chatbot created has a high hardware usage, as running an LLM locally is resource intensive. This was mitigated through the careful selection between the size of the chatbot and the research that was being carried out. By the use of software and having to install the LLM used locally, this made it so the research could only be ran locally to hardware after all requirements had been installed. Creating a balance between creating a smarter and more state-of-the-art chatbot compared to the limit of how it could be used and applied.

B. Design & Development

To create this chatbot, open source software and LLM models were used. Within this research, it was decided that Ollama and Deepseek-r1:8b would be used. These are both open source allowing for them to be easily used for research purposes. Ollama was created to allow LLM models to be ran easily and locally, making it ideal for research purposes due to it giving greater control over privacy, security and how easily it can be modified. The Deepseek LLM is a reasoning model, meaning it uses logical processes to generate an output as well as keep track of some conversation history. Due to this research artifact being run locally, the memory function of the model was disabled as each reply added context for

the chatbot to consider before replying, increasing the amount of time to reply per message. Utilising a trained LLM and open source software reduces the time needed within this study allowing for it to fit within the short time frame that has been given. This allowed for all elements of a research study to be carried out. Ollama is a stand-alone software, but it does have the ability to be integrated into Python scripts using libraries which was the method used for this study. Since this was developed in Python, tkinter was used to create a user interface that participants could easily use and understand. This remained simple to not confuse participants or introduce unnecessary features to the research. The interface consisted of a text box the participant could type into, a send button and a message log that showed the chatbot responses and previously sent messages.

Using a combination of Ollama, Deepseek and python scripting the desired artefact for this research was created. Ollama and Deepseek were responsible for creating and managing the AI chatbot and the data associated with it to create replies to the participants' questions. The Python programming handled the front-end user interaction, passing data from the user to the chatbot, which ensured responses were easy to read and the prompt engineering portion. An agile workflow was followed throughout the development of this artefact, with functionality being added and then tested to ensure consistent progress with minimal bugs being introduced. This follows the fundamentals of the Booch Method.[35] Bugs that were found during this process were fixed before the next stage of development could begin to ensure that no knock-on effects could occur by allowing bugs to make it into the production pipeline. Once the Deepseek-based chatbot was functional the prompt engineering began to add the personality to ensure a good foundation and so that clear developmental stages could be used.

C. Implementation

When the development and testing had completed, the participant and data collection could begin. This involved allowing participants to talk to the chatbot and ask it questions. The participants were given the scenario of meeting someone for the first time to act out with the chatbot. They then answer a questionnaire asking them to make assumptions about the personality that they perceived from the chatbot. Due to the limitations of using a locally run LLM for the chatbot this meant that it could not be packaged easily. This limitation also meant that it could only be run on machines that had all the necessary software and the correct model installed. Participants were still able to successfully interact with the chatbot on hardware that had all the relevant software and data already installed. During the implementation and collection phase, it was made sure that the chatbot would not remember any information that was entered into it. This was to stop any personal data that the participant could have entered from being stored. Not allowing the chatbot to form memories ensured that each participant would talk to the same personality, and nothing said within the previous conversation would affect the overall experiment.

D. Maintenance

As the artefact was used within a study, things like how the chatbot ran, the prompt effecting the personality or the model could not be changed. This is due to not wanting to invalidate data that had already been collected. The maintenance phase consisted of continuing checks on the chatbot to ensure it was behaving the same way throughout the study. This ensured that things like the user interface continued to work as expected while not allowing the data collected to be irrelevant to the study. Throughout development, version control was used. This ensured that changes could be saved online as the code was being altered. This also leaves a log of changes that have been made, meaning any issues a previous commit could be rolled back to, ensuring a bug could not break all the code. This also ensured that there was a log of code refactoring as it can be easily seen that the code went through many iterations until it worked to a good level. Throughout the code there are comments explaining what functions and lines do so that future research can be easily done using the same system. Documentation is within the GitHub repository, explaining in the readme how to download dependencies and ensure it runs as intended.

VI. QUALITY ASSURANCE

Testing is an important part of any good research project. It ensures that quality is kept consistent. Testing ensures that the chatbot responses are accurate, helps to find errors early, makes sure it works with external services and more. Ensuring consistency within this project was important due to the use and collection of data from participants. Since the data collected was based on the opinions of the participants, if the chatbot did not behave or perform consistently, this would cause skews in the data. Within this study unit testing, AB testing and mock user testing were performed. To keep on-top of bugs within the project bug tracking was used within the GitHub commit notes. This was noted in a place that was easy to access and allowed easily tracking of found bugs.

A. Unit Testing

Unit testing was used to ensure that the core functionality of the chatbot was reliable and worked as intended for this research. Pytest was used within this project to create reliable and simple unit tests. This testing framework made it easy to create unit tests throughout development. These tests could be grouped into different scripts inside a test folder. Pytest would then handle the finding and executing these tests, giving an output of any tests that failed and saying which tests passed. To write these each test was defined as a function with the test interacting with the function being tested and then taking the value it is expecting and comparing the two. The first function that was unit tested was the response cleaning function. This function was vital for ensuring readability for participants. This test ensured that all thinking logic would be removed no matter where it was within the text, it also ensured that if there were no think tags, it would not remove text that was important to output. It was also important to check if the output was empty it did not insert text, as this would

cause many unforeseen issues if text was being inserted from outside the chatbots LLM. The second script of unit test makes sure that the LLM model is properly invoked and is giving replies through the Python chatbot script. This involves feeding predetermined questions and ensuring that the prompt is added to the chatbot, that responses contain strings greater than zero, and that it can handle inputs of any type of characters.

B. A/B Testing

A/B testing was used for the prompt engineering development portion of this project. This consisted of asking two chatbots with differing prompts the same questions to ensure correct personality identifiers. These questions were designed to be specific for each personality trait to ensure that answers were clear to understand if the prompt had correctly been taken in by the bot. Through running these two bots with slightly different prompts and finding the one with the better responses development could be done quickly. Once the better prompt was found, it could be altered further to correct questions it responded to incorrectly or parts where the other chatbot may have been better. After using this test repeatedly on different iterations a bot that had a strong alignment to the desired personality required for this study was formed.

C. Impersonating User Testing

Once all other testing had been carried out and the artefact was ready for deployment among participants, users tests could be used. In the interest of time and ethical concerns, mock user tests were used where conversations a user might have were imitated. This allowed for vital information as participant time was not wasted due to attempted data collection finding issues. During these tests, it was found that users did not know what type of questions to ask the chatbot and giving them a too open-ended scenario led to the user not getting enough information. The result of these conversations is that the chatbot would respond in an undesirable way. From this test, the example questions and the participant scenario was created. This gave the participants a few questions as examples they could use and adapt to think of their own questions while also giving them context that was within the information sheet but needed repeating to ensure they consciously knew the scenario they were in.

VII. RESEARCH METHODS

A. Philosophy

Within this study, the Saunders Research Onion[36] was used to find a research philosophy that fits into the scope of this study. This study used a *pragmatic* research philosophy. This worked well as it looked at using the best fit and most up-to-date method for looking into the study. This allowed for the state of the art to be used while staying within the confines of this study. While allowing for adjustments to be made throughout the process to ensure timely and accurate development. This allowed for the ability to scale the research by allowing the creation of many different chatbot personalities by just allowing for the prompt to be altered to

create a different persona. This allows for the personality traits that are exaggerated and omitted to be easily interchanged so this research can run in many different contexts.

Within this study, the *deductivism* method was used to generate hypotheses and thoughts before the experiment was carried out. This allowed for the data to prove and disprove hypotheses and allowed for the research to guide future work through its findings. This study followed the mono method approach where only one type of data was collected. This data consisted of what personalities the participant can tell the chatbot has. This singular data collection point is then used to graph and visualise the resulting outcome. By using *grounded theory*, the collected data guides the study and what the results show from it. This way of thinking lends itself well to this study where there has not been a lot of literature found on this specific topic, making the data gathered show the way for future works and making research opportunities for the future. The data collected in this study is *cross-sectional* as it is measured from the one point in time when the data was collected. As technology improves and humans continue advancing, the results that make up research may no longer be correct as people could change how they view chatbots. Leading to this study being correct for the current state of this field.

VIII. ETHICS AND LEGAL

There were many legal issues to be considered within the study and creation of chatbots and conversational AI.[37] Intellectual property was considered, as the creation and data used to train the model can come from different places. The training data needs to be gathered from a reputable source to ensure that the bot comes across as human and is ethical to use within a study. With all possible data, there is a chance it is collected illegitimately due to someone not checking the copyright of the data or disclosing the use during collection. It was important within this project to ensure that any data that was added or collected within this study was done within the rules of law and ethics. Human rights protections were considered as the data was collected from willing participants. Chatbots can take in and learn personal information that is protected characteristics. This goes against the right to identity, equality of rights and possibly child protection due to a child not being able to recognise the difference between a chatbot and a human. Within this study children were not used as participants, so this was not an issue but is still important to consider with academic studies. To maintain that the bot remained within all laws and rights, it was tested regularly, according to the test plan, to eliminate any rogue or unwanted behaviour. Data was not collected to train the chatbot, as a pre-trained model was used to allow for state-of-the-art technology to be used. The responsibility of checking data and ensuring it is fair use is the responsibility of the creator. Ensuring the model was open source and was allowed to be used within a research study was important to ensure it was legal. It was important to follow the guidelines for participant participation as the HEXACO Personality Inventory specifies, "Please note that we do not allow researchers to collect data through

an online survey site or an app accessible by the general public. Those online survey pages or apps must either be password-protected or not searchable through search engines.” This allows for researchers to use this freely as long as it is password protected, to comply with data collection guidelines. As the project collected data from participants, it followed General Data Protection Regulation (GDPR) and the UK’s data protection standard.[38] This act dictates how data should be handled, stored and used. Information was given to participants in the form of a data sheet on how their data was used when they agreed to be a part of the study. Only data that was needed to complete the study was collected. This data was stored in a secure place, which was the university provided one drive. This data was deleted once the experiment had concluded and all graphs and needed data had been transferred into this paper. This participant data was stored anonymously. Participants were given the needed and correct information and checks to ensure they were willing to participate according to ethic guidelines, but once they had engaged with the study, all their data was anonymised so it all looked the same with nothing tying it back to the participant. This made it impossible for any data to be removed since there was no way to tell what data came from a specific individual. The participants had the right to withdraw at any time during the study to follow ethical guidelines, even though the data was impossible to identify and remove. The Nuremberg Code[39] and the Declaration of Helsinki[40] state that all participants in a study should have informed consent, have a scientific purpose, avoiding harm and giving participants to withdraw from a study with no consequence. As this study involves participants, this was followed very closely to ensure that everything was done to a respectable standard and no harm to reputation could be caused.

Chatbots are used in things like websites to deal with customer questions. This takes away the need and wait time it takes for an employee to reply to these messages, making the customer experience more effective. This has a negative impact of taking away jobs that could be filled by a human. Users and participants were made clear on the abilities to build a sense of trust and clarity for the experiment. The digital divide was also considered, as any benefits that chatbots and conversational AI can achieve will be only for the people who have the technology to access them.

While working with chatbot and AI models of the like, a level of human input was considered within the project. As Siau and Wang found in their study[41] is that human based errors and ethical issues can make their way into a technology like this. It was important to test to ensure these were not present within the experiment. Data concerns are also talked about where large amounts of data are used to train large language models (LLMs), AI tools and more. The training data needed to be checked to ensure that it is legally right to use and checked for sensitive or protected data. Since the data used to train bots and models are human-made or collected from the internet, there was a possibility of human bias. These biases can consist of things like stereotyping, selective perception, the bandwagon effect and more.[42] These affect and skew the outputs of a model, possibly making them unsavoury to use.

This point was countered by testing and by using a pre-made LLM model. As it had been produced by a company as an open source project it required to be ethically and legally tested to be released and used within the public domain. Sengupta, Garg, Choudhury and Aggarwal suggest the usage of machine ethics. As defined by Anderson and Anderson, ”machine ethics is concerned with how machines behave toward human users and other machines” [43] While not all machines may have an ethical impact, they should still be considered with ethics in mind.[44] Machine ethics was good to keep in mind throughout all parts of the research, development, testing and the data-gathering portion of a project like this. The usage of external testing and risk assessments were perfect to use in situations like this to counter and reduce any ethical issues that might occur.

To maintain professional standards the British Computing Society (BCS) code of conduct was followed throughout this study.[45] This code focuses on acting in the best interest for the general population, keeping all work at a high standard of quality, perform at the best of their abilities and to uphold reputation and professional development of computing subjects.

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IX. SCIENTIFIC HYPOTHESIS TESTING

To test the hypotheses laid out within this paper, participants were gathered and allowed to talk to the chatbot. They asked the chatbot introductory questions and what could be considered first date questions to get a sense of personality.

A. Data Gathering

Microsoft Forms was used to collect the data from participants. This allowed for easy data collection as it could be filled out straight after having a conversation with the chatbot without needing to interact with anyone. This avoided the possibility of any bias entering the form that could have skewed the data one way or another, which would have caused issues within the statistical analysis. Within the Microsoft Form, the only data that was gathered was the data relating to the study. Any personally identifiable information was excluded to make sure that data regulations and laws can be complied with as easily as possible. This ensured that the data was anonymised so certain data regulations and laws would not apply.

Microsoft Forms was chosen due to it being an industry standard and being created by a reputable company. This made it perfect to use as it could be easily accessed to gather the data to be used within other software. It could also be very easily accessed and shared with the relevant parties and the participants. Microsoft Forms is a secure data platform, which made it better suited for this research. Microsoft Forms follows data protection guidelines and, as a third-party source falls under regulations for data management that make them a secure choice.

²<https://learningspace.falmouth.ac.uk/mod/quiz/review.php?attempt=26290&cmid=408100>

B. Data Management

Throughout the duration of the data gathering portion of this project the data stayed within the Microsoft Form and the Microsoft servers which are secured through password protection and two-factor authentication. Once the data gathering portion was completed, the data was kept stored within Microsoft OneDrive, which falls under the same protections as Microsoft Forms.

The data was exported from the forms to a CSV file. This format is supported by lots of software, so it is easy to import data into it and export data for use. This file was stored securely in the same places as stated above to ensure ethical, professional and legal guidelines were followed. After marking the participants' form data, changing it from the Likert scale answers to numbered scores going from 1 to 5 according to the HEXACO marker scheme.[REF] The data could then be used from within the CSV file to calculate the mean for each personality trait. The mean value of each trait was then used within RStudio to perform statistical tests.

C. Data Analysis

For the data analysis, a one sample t-test was used to provide an evaluation of the mean rating of the participants compared to the designed personality. The chatbot personality was the control variable in this study, and the participants' ratings of traits were the dependent variable being measured. The independent variable was the different participants as each person has a different perception. The Likert scale was assigned 5 measurements from strongly disagree to strongly agree. This follows the HEXACO personality test, and then the mark scheme can be used to translate the answers to numerical scores. This follows strongly disagree being worth 1 point to the opposite end of the scale of strongly agree being worth 5. Some questions had a reverse scoring key so the scale was reversed, with strongly disagree being worth 5 to strongly agreeing being worth 1. The mean of each trait can then be used to see if the average participant can accurately tell the personality.

The G*Power software [46] [47] has been used to calculate the sample size of participants that is needed for this study. The effect size has been chosen to be 0.5, which is a medium effect for this study. This reflects the relationship between the variables of the chatbots' personality and the users' perception. This has been chosen due to the participant's opinions on personality will be influenced by previous knowledge with what they associate with each personality trait and the fact that each participant can interpret the same types of messages in different ways. The effect size has not chosen to be smaller due to the personalities will be directly affecting participants perceptions and be the primary focus of the study. The alpha error rate is 0.05 to indicate a type one error will be accepted 1/20 times. This means there is a 5% chance of the null hypothesis being rejected when it is true. The type two error rate is set to be 0.8, meaning that there is an 80% chance of rejecting the null hypothesis if the participants can recognise personality traits. The test is set to be two-tailed since there is

no predetermined direction, and participants could incorrectly assign personality traits in either direction.

RStudio was then used to analyse the data. This performed statistical tests and generated graphs for visualisation. Statistical significance was calculated by the use of t-test to see if the collected data meets the hypotheses. This test ensures that the findings are statistically valid and calculates the error rate to ensure it is acceptable. R Studio was used for this as it is a statistical-focused software that is ideal for this type of data and tests. A t-test was chosen due to the focus on measuring the difference between a known value and a group value. Within this experiment, that was the set chatbot personality value and the participants' average opinion.

X. RESULTS & ANALYSIS

Personality Traits	Hypothesised Value	Participant Average
Honesty-Humility 1	3	4.1
Emotionality	3	2.6
Extraversion	1	3.3
Agreeableness	5	3.8
Conscientiousness	5	4.1
Openness to Experience	1	3.7
Overall Personality	3	3.6

TABLE II
THE RESULT VALUES FOR EACH PERSONALITY TRAIT AGAINST THE
THEORISED VALUES

The that had been collected from the study has been formatted for display in Table 1 above. This data shows the average value across all participants for each personality trait within the HEXACO model. It also shows the overall average calculated from each traits average score.

To create graphs and perform statistical tests on the collected data, RStudio 2024.09.1+394 was used. The libraries tidyverse and ggplot were used to collate and graph the data, see the Appendix for the code used. Figure 1 shows the results that were collected against the hypothesised values that the chatbot was set at. This visualisation of the data shows the trend within the participants of what they thought of each personality. The box and whisker plot was used for this ability to show the data as it is easy to see where the quartiles are.

Since the nature of this test, the statistical tests performed on the collected data were one-sample t-tests. This is so that it can be determined if the unknown mean is different from the hypothesised value, which was formed from what personality the chatbot had been set with. If there is a small p-value, that means the participants' observations are different from what the chatbot personality was set to.

A. Hypothesis 1 - Participants can recognise Agreeableness in the chatbot

This hypothesis was for Agreeableness of the HEXACO personality traits and if the participants could recognise it within a conversation with an AI chatbot. Figure 1 shows that the median and interquartile range trended towards being -1 points away from the hypothesised value. A t-test was carried out to analyse the difference between the reported values and

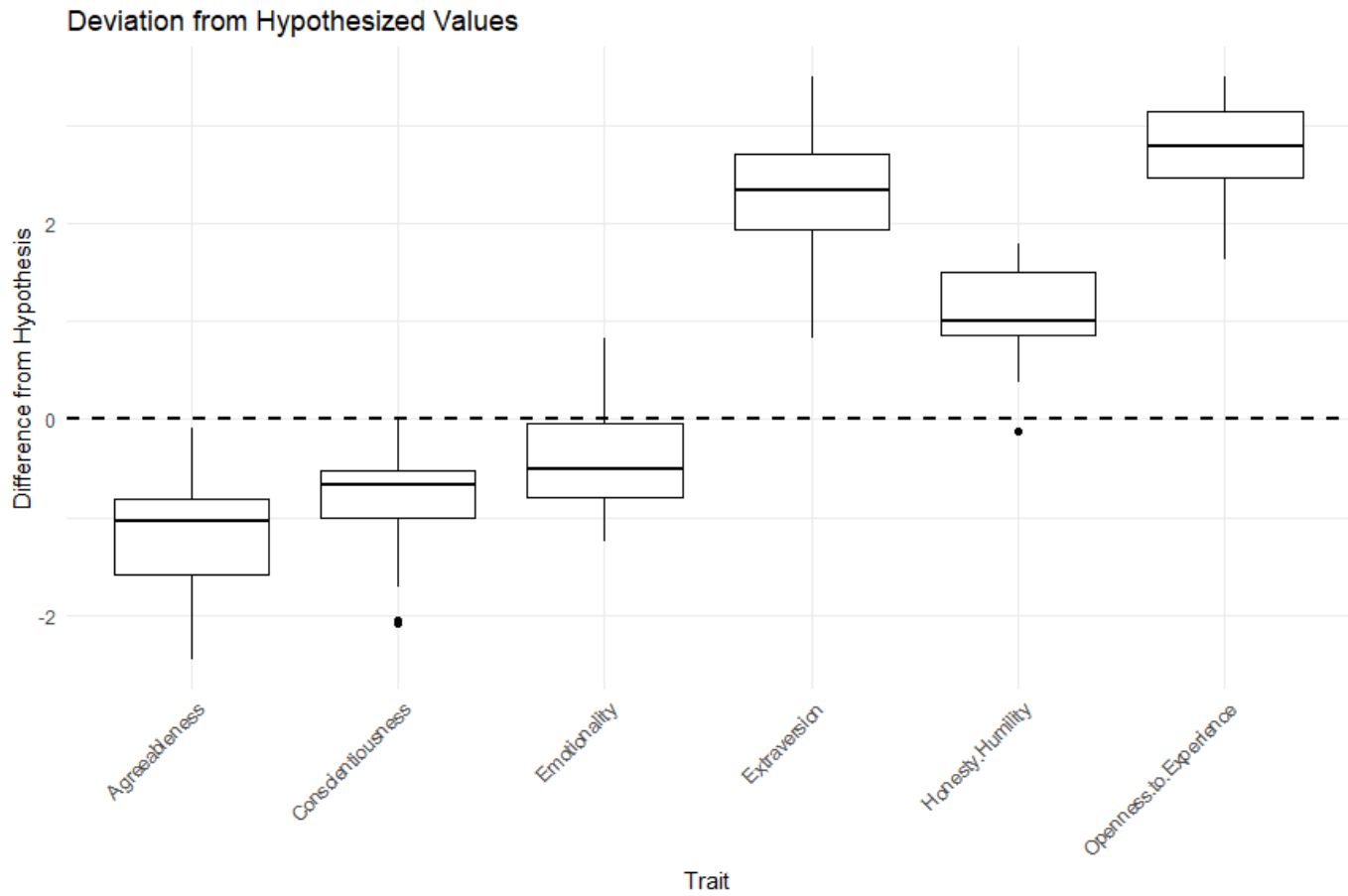


Fig. 1. A vertical box-and-whisker plot, showing the distance between the collected values and the hypothesised values

the set value. This test had a p-value of $p = 4.436e-9$ and a t-value of $t = -9.301$. The p-value is very low, rejecting the null hypothesis. The mean value of the differences was 3.778986.

B. Hypothesis 2 - Participants can recognise Conscientiousness in the chatbot

The focus of this hypothesis was to see if participants could recognise Conscientiousness within the chatbot. From figure 1 it can be seen that the median is approximately -0.6 away from the hypothesised value. From the t-test the p-value was $p = 2.84e-7$ with a t-value of $t = -7.2606$. With a p-value being low like this it rejects the null hypothesis, meaning that participants could not correctly identify conscientiousness. The mean value within this is 4.188406.

C. Hypothesis 3 - Participants can recognise Emotionality in the chatbot

This hypothesis focused on Emotionality. Figure 1 shows that the median was around -0.5 away from the hypothesis, but the hypothesised value was within the outer quartiles. This had a p-value of $p = 0.002844$ and a t-value of $t = -3.3578$. As this p-value is below 0.05 it suggests the null hypothesis can be rejected and that participants could not recognise emotionality. The mean value was 2.597826.

D. Hypothesis 4 - Participants can recognise Extraversion in the chatbot

Hypothesis 4 is based on the observation of Extraversion. Figure 1 shows that the median of the plot was approximately +2.4 from the hypothesised value. The t-value of this was $t = 18.313$, and the p-value was $p = 8.337e-15$. The null hypothesis can be rejected with a value this low and suggests that participants cannot recognise extraversion. The mean value for this was 3.318841.

E. Hypothesis 5 - Participants can recognise Honesty-Humility in the chatbot

The participants recognition of Honesty-Humility was questioned within this hypothesis. The median was +1 away from the hypothesised value as shown by figure 1. The p-value for this trait was $p = 1.992e-10$ while the t-value was $t = 11.024$. With this p-value the null hypothesis is rejected so it indicated that the participants could not recognise honesty-humility.

F. Hypothesis 6 - Participants can recognise Openness to Experience in the chatbot

From figure 1 it can be seen that the median was approximately around +2.7 away from the hypothesised value. This trait had a t-value of $t = 24.125$ with a p-value of less than $2.2e-16$. This is the smallest p-value within this study and with this

it suggests that openness to experience cannot be recognised by the participants and the null hypothesis is rejected. The mean for this trait was 3.735507.

G. Hypothesis 7 - Participants can recognise the personality on average in the chatbot

This hypothesis was for the overall personality trait of the chatbot, so it uses the data gathered for each of the other hypotheses to judge if the participants could tell the personality as a whole. From Table 1, we can see that the average value was 3.6, and the hypothesised value was 3, so the overall score was +0.6 away from the hypothesised value. It had a p-value of $p = 2.562e-8$ and a t-value of $t = 8.4075$. With a low p-value it suggests that the null hypothesis can be rejected. The mean value for the overall traits was 3.61564.

XI. DISCUSSION

The results from this study suggest that users cannot recognise traits presented by a chatbot due to all the hypotheses being rejected. The data that has been reviewed looks at the average data from 23 participants and what they thought of the chatbot. This section focuses on the potential reasons behind this. Within Figure 1 the data is displayed within a box and whisker plot. This shows the data of the participants and the averages within this group. This shows that none of the data has an interquartile range over the hypothesis line, which indicates that there was not a strong trend between the participants and chatbot recognition. Looking back to the research question, *How accurately do participants identify HEXACO personality traits from a personality-focused chatbot?*, the data shows that users do not correctly identify these set personality traits. From figure 1 it can be seen that the closest personality to the set value was emotionality. This shows that the participants could most accurately tell the amount of emotionality within the chatbot.

Within previous research, OCEAN personality traits have been reviewed within chatbots and the Agreeableness trait of the HEXACO model has been reviewed. In their paper [28], Yorita et al. created and reviewed two facets of the OCEAN personality model. This paper strives to prove a link between the recognition of extraversion and agreeableness. This paper shows that chatbots that adapt how they act to suit the user get given a higher score in the support that it gives them. The research from this paper disagrees with what has been found within this study. Within this study, hypotheses 1 and 4 have been rejected, meaning that the participants could not recognise these personality traits.

In their paper [32], Kuhail et al. researched the impact that personality has on human-chatbot relations. Using the 5-factor OCEAN model, they found that extroverted participants had a higher rating of support and trust with extroverted chatbots. Agreeable and conscientious participants had inconclusive results meaning that they made no reasonable difference or could not be recognised by the participants within the paper. Within their paper they had a focus on chatbots involved within academic advising systems and from their study found that there is a correlation between similarity between personalities

and an increase in trust. From this paper, we can infer that the participants were able to, at least subconsciously, tell what personality each chatbot had as they got along better. From the research findings of this paper, it was found that participants could not accurately recognise the personality traits shown.

From these two papers re-examined against what has been found within this study and paper, there is a clear divide in opinion. Since this study uses the HEXACO personality model, which measures an extra measure of personality, it could be that users cannot accurately measure against that. Within the other papers, there were select personality traits focused on. This focus might have led to more developmental time to focus on the personality traits being tested to expand upon them and ensure that they have been made in great detail. With this paper, using the HEXACO personality was a new focus as the other papers focused on the older 5-factor OCEAN model. Within this study a chatbot was built upon existing AI framework and LLMs. This study implemented personality over an install of DeepSeek. The other papers created a chatbot from the ground up with personality in mind. This difference could make the other studies' chatbots have a more ingrained personality, making it more accurate in displaying personalities.

Within this study, the sample size that was used was smaller than expected. G*Power recommended a sample size of 34. Within this study, only 23 participants were collected, making this data unreliable and making this study less reliable. Having a smaller data size than recommended can cause an increased risk of errors within the results and skewing data due to anomalies. This difference between the results and what was expected from the literature could have caused the difference that is seen. The data collection method used could also have skewed the data. While it is based on a tested psychological model, it asks for participants to make assumptions or guesses. This uncertainty could have shifted the results and ratings.

If these findings are accurate within further research then it prompts forwards further looks into the relationship between human and conversational agents. From the findings here using the six factor model increases the data that we can see as all the traits are at different levels. This study also displays how users can vary within opinions of how they perceive personality through conversation, with some participants ranking much differently from the consensus. The findings also suggest further research as the emotionality traits box and whisker plot in Figure 1 intersects with the line of expected value. This implied that some users could recognise the trait, this data suggests further research, possibly with more participants and a further developed chatbot.

While the results are not statistically conclusive, there are clear traits that are more closely recognised than others. This could affect practical applications by development of these having a focus on these personality traits as users possibly have an easier time recognising these personalities. Due to the sample size not meeting criteria no practice or policy changes should be made of this data until it is researched further and should be interpreted with caution. If similar findings are found within a larger study that better represents the research and data that would need to be collected, support,

retail, and mental health chatbots could have less of a focus on personality development. Allowing for further research on how to make them more effective, such as quicker response times and affects of more training resources.

XII. LIMITATIONS

A. Participant Numbers

One of the main limitations within this study was the inability to collect enough participant data to meet the recommended value calculated within the G*Power software. This limit affected the statistical power analysis, increasing the chance of type II errors. This means that these findings are limited as an effect may have failed to be detected when one exists. If the recommended participant number was collected these results could be considered more reliable and indicative within the research.

B. Users Perception of Personality

Another limitation within this study was using the HEXACO personality model. This model was good to use due to it being the most up to date and other papers not making use of this model. The problem with it within this study is that participants must understand it. Due to the nature of collecting results, the participants had to answer a series of 60 questions. The 60-question form is the shortest one, to get more accurate results the 100 or even 200-question version to ensure more accurate measures of the personality. There is also the issue of how well participants can truly gauge personality from a purely text-based interaction. If a larger question number was used, each personality trait would have more participant data to collect, creating a more accurate visualisation. This study could not achieve that due to the time constraints and participant time, as many were bored after filling out the 60 questions. To fix the issue highlighted with a purely text-based interaction, visual cues like a changing face or making use of text to speech.

C. Hardware Limits

A hardware limitation was present within this study. Running the LLM model locally to generate the chatbot replies used the hardware inside the devices that were accessible. These devices consisted of high-end user components. This limited what the study could achieve by reducing the size of the LLM that could be used. This is due to the high computational power that generative AI uses. Using a computer or server that is dedicated to running AI models would allow this study to achieve better results. Better hardware would allow for a larger size model, making it more intelligent, and the hardware would allow for quicker response generation.

D. Time Constraints

The main constraint within this study has been the time limit. Due to this time limit, the scope of the study had to remain small to ensure that it could all be completed to a good quality in time. Without this time limit, there would be more time to research, develop and collect participant

data. This would improve the experiment by having a better made chatbot model, which would possibly display personality better, ensuring more accurate responses from the participants. Additionally, a longer time limit would have allowed for further participant data collection without having to cut off the data collection so quickly. In future research a longer amount of time within this study should be considered.

XIII. CONCLUSION

This study has researched if participants could accurately recognise HEXACO personality traits from an AI based chatbot through a conversation. Through the analysis of the participant responses the results suggest that participants were unable to accurately identify the intended personality traits within the chatbot. The findings from this emphasise potential limitations on how personality can be conveyed through a purely text-based interaction and question how important it is to make a chatbot with distinct personality traits. Consideration of this is imperative to improve human and AI communication.

A. Future Work

This study found challenges in recognising personality traits from a chatbot, limitations may have influenced participant responses. Further research could explore the effect of having facial expressions or tone of voice controlled by the chatbot could improve the perception and recognition of personalities. By making use of text-to-speech and character face models with animations, these could be easily simulated for experimental design.

Using more advanced and larger AI LLM models could provide better insight and more accurate data. By making use of better equipment that can run larger models at a fast rate and adapting these larger models better through the use of training, a more accurate personality could be made. This could lead to participants being more confident in their perceptions. Further studies may also focus on having participants interact with the chatbots for a longer period. The use of extended conversations may allow for better trait recognition as participants will be able to ask more questions. Exploring these areas will develop and contribute to more human chatbots that have better engagement.

B. Pathways to Impact

Through the findings of this study insight into the development of personality based chatbots. AI powered tools and applications have increasingly been created and integrated into many different software throughout peoples daily lives. These applications are things like customer support, assistants and mental health services. Within these examples the ability to perceive a chatbot as more human is important to trust.

The results suggest that users cannot currently recognise specific personality traits which could impact engagement, trust and willingness to use a chatbot. Alternative methods should be considered to enhance these conversations to ensure that an uncanny effect does not affect users. By making use of additional technologies such as text-to-speech, using adapting

models to change conversational styles or making use of facial expressions. These additions and improvements to the overall aspect of personality conveyance could improve recognition, making interactions more natural.

The long term impact of expressive AI chatbots could lead to more personalised services, benefitting things like mental health services and virtual assistants. If personality can be recognised by users, then user confidence and satisfaction could increase. This would mean a higher level of support would be a viable option to give through an AI chatbot. The possible increase in trust could improve mental health services massively as there would be an always available chat option that can help people through bad periods of mental health and direct them to services they need without losing the trust of the user. Addressing and researching further what was found within this paper is essential for a balanced service that can better mimic human interaction.

APPENDIX A REFLECTIVE ADDENDUM

I am proud of the research that has been carried out, but there are flaws and considerations that, if addressed early on, could have led to a better project. This project has been excellent for learning lots of different aspects of technology. Before starting this, I had a surface-level understanding of the technology and I can now comfortably say that my knowledge has deepened. I know there is much more I can learn. This section critically reflects on the time and issues I faced throughout this project and what went wrong. This project affects my future work by allowing me to understand modern systems better and pay closer attention to the current focus of the tech industry, which is generative AI. Throughout this project, multiple challenges were encountered. The six most prominent ones that affected me the most have been chosen to reflect on so that improvements can be made for future projects.

A. Cognitive - Lack of Knowledge in the Field

Throughout my studies, I had not encountered technology like this causing a gap in knowledge that was not considered while creating a proposal. Changing the project from what was outlined in the proposal after the Christmas break created a bigger gap in knowledge, as what I had researched during the first term was not as relevant. This significant knowledge gap caused me to spend lots of time at the beginning of the study block researching how what I set out to complete could be achieved. Through many days spent researching everything I could, I managed to find a development process that was feasible. For future projects, any knowledge or skills that are needed to complete it before trying to start development. Allowing myself time to practise these skills and have a strong knowledge base. This would be done before starting the project or within the first few weeks, ensuring the correct software and languages were selected and learning what is needed to complete the whole project smoothly. To improve this using a smart target I would have spent more time learning relevant knowledge, taking the first few weeks to read papers and

relevant guides. Since I will have more knowledge of the field it will improve my outcome.

B. Procedural - Fine Tuning and Prompting

Using a LLM was a risk taken within this project to conserve time and to create something that utilised state of the art that could help develop skills useful within industry. DeepSeek was used since it was open-source, newly released and fulfilled what was needed for completion of this project. This project required good knowledge of how AI and LLMs work in order to achieve a good result. Since I did not have enough knowledge within this field this challenge was more difficult than expected. This was used within the project due to it being the best way to customise an already existing model to change its behaviour to what is needed.

This issue affected my output due to it being a complicated process to understand and carry out. This issue with applying these techniques within the project caused delays as testing had to go on for longer than expected as issues kept appearing from bugs I was making. To set a SMART target for this, I would look further into unique scenarios, ensuring I fully understand them before implementation. Watching and reading tutorials and spending a few hours a week to build a better customised chatbot within the first 3 weeks of the project would have made development a lot easier.

C. Affective - Frustration with the Testing Process

During the testing process, I experienced high levels of frustration and stress due to not understanding what needed to be done. This also carried through to a lot of the rest of the project relating to the testing process, as I was consistently unsure if what I had done was correct even after assurance from peers and supervisors. Testing is an important process within any project to ensure it works correctly. Due to not having done much formal testing prior to this module experiencing for the first time while under pressure from the module already caused more stress than what was required. This affected my output as I could not properly work on the tests. Every time I worked on these tests, I was stressed and unsure, creating a less than ideal working environment due to not properly understanding and the feelings of stress just collating. I eventually got through these feelings to complete the testing phase, but looking back, it could have been finished a lot earlier and with less frustration if I had dealt with my emotions better. By controlling my emotions by not allowing myself to stay stressed at the project and by ensuring sufficient breaks are taken. By doing this there would be a smoother workflow that would have daily mental health checks.

D. Dispositional - Procrastination

Throughout this project I dealt with lots of procrastination. This led to me not wanting to work on the project as much as I should have or could have. This led to me getting behind and wasting time when I did not need to as nothing was preventing me apart from the lack of motivation to work. This affected the output by slowing down all parts of development

as I would always find other things to do rather than focus on working. This reduced the consistent output that I could have had throughout the project, which would have left me in a more comfortable position to finish the project long before the deadline to check over the project and paper properly. To set a SMART goal to improve this, I would make use of a routine working schedule, spending 3 hours each day doing work and nothing else. This would allow for better workflow while giving time each day, I will implement this in my other projects moving forward, checking each week that I completed enough work each day.

E. Interpersonal - Interacting with Participants

I had difficulty approaching and talking to potential participants within the study. While trying to find participants to collect data from, I experienced challenges going up to anyone to ask them to participate in this study. This was due to asking some participants and them saying no, leading me to automatically expect lots of other participants to deny taking part in the study. It was necessary to gain participants within this study to collate and collect data needed to generate results to analyse against the hypotheses. This affected my output by putting my data collection portion a week behind, causing me to take longer than expected while gaining fewer participants than needed. This affected the output because I was not able to write any of the results, analysis or conclusion while attempting to get more results as after looking at the results, it would not be scientifically ethical to continue collecting results due to the chance of me influencing them in any way. To deal with this, a SMART target can be formed. Using a participant recruitment script, five participants can be approached a day, tracking successful participants and adjusting techniques if needed. A script can be created to introduce the study and answer common questions, this goal will allow recruiting of participants over two weeks to get enough to meet required values.

F. Practice - Chatbot Response Time

While creating the chatbot, I faced issues with the response time of the AI chatbot. These issues stemmed from using too large a model so the decisions took much longer than expected on the hardware that was being used. This combined with using a reasoning model, made the response times a lot longer until a smaller model was used. Chatbots must take time to generate a response that matches the question asked. If a larger model is used, the answer tends to be better, but it takes longer to sort through the internal data to generate the response. This slowed down my work as I thought it was an issue that I was causing through my use of coding, so I was stuck fixing a non-existent bug. This took away time that could have been spent working on other parts of the project that would have been much more effective to the final artefact that was developed. To improve response time smaller models should be used first. The response time should then be benchmarked each week as changes are made against models of larger sizes so that response time versus accuracy can be mapped. This ensures both efficiency and quality are kept to a high quality while not losing development time.

APPENDIX B TESTING ADDENDUM

Model Response Unit Tests

```
import unittest
import unicodedata # Used for normalising unicode data for
↳ comparisons between strings.
from langchain_ollama import OllamaLLM # type: ignore
from langchain_core.prompts import ChatPromptTemplate #
↳ type: ignore

class TestPromptModel(unittest.TestCase):

    #Init model and prompt before each test to ensure no
    ↳ cross over. Must be called setUp to interact with
    ↳ the unittest framework.
    def setUp(self):
        self.template = """
        Respond to the user's question below.

        Question: {question}
        """
        self.prompt =
        ↳ ChatPromptTemplate.from_template(self.template)
        self.model = OllamaLLM(model="deepseek-r1:8b",
        ↳ temperature=0, seed=42)
        self.chain = self.prompt | self.model

    #Ensure the question gets correctly inserted into the
    ↳ prompt so that the chatbot can clearly see what
    ↳ the input is.
    def test_prompt(self):
        formatted_prompt =
        ↳ self.prompt.format(question="Prompt
        ↳ Inserted?")
        self.assertIn("Question: Prompt Inserted?",
        ↳ formatted_prompt)

    #Ensures that there is a response is not empty and has
    ↳ a reply. Checks it is in the format of a string
    ↳ and it is more than length 0
    def test_botInvokes(self):
        response = self.chain.invoke({"question": "What
        ↳ day is it?"})
        self.assertIsInstance(response, str)
        self.assertGreater(len(response), 0)

    #Checks if the bot responds to an empty or incomplete
    ↳ input
    def test_botEmptyInput(self):
        response = self.chain.invoke({"question": ""})
        self.assertTrue(response.strip(), "Bot should
        ↳ reply to an empty input.")

    #Checks if the bot replies to long inputs, eg 1000
    ↳ words
    def test_botLongInput(self):
        long_input = "hello world" * 1000 # 1000
        ↳ repetitions of "AI "
        response = self.chain.invoke({"question":
        ↳ long_input})
        self.assertTrue(response.strip(), "Bot did not
        ↳ failed at long input.")

    #Checks to see if special characters are properly
    ↳ picked up
    def test_botSpecialCharacters(self):
        special_input = "@##$%!!*%(%) "
        response = self.chain.invoke({"question":
        ↳ special_input})
        self.assertTrue(response.strip(), "Bot failed at
        ↳ special characters.")

if __name__ == "__main__":
    unittest.main()
```

Response Cleaning Unit Tests

```

import unittest
from ChatbotLLM import cleanResponse

class TestCleanResponse(unittest.TestCase):
    #If think tags are used for chain of logic reasoning
    ↪ then ensure they are removed to make sure user
    ↪ only gets the relevant reply.
    def test_removeThinkTags(self):
        botResponse = "<think>Internal reasoning for
        ↪ chatbot logic that determines reply</think>
        ↪ Hello, how are you doing?"
        expectedResult = "Hello, how are you doing?"
        self.assertEqual(cleanResponse(botResponse),
        ↪ expectedResult)

    #If think tags have not been used since there is no
    ↪ reasoning used then the response should remain
    ↪ unchanged after going through this
    def test_noThinkTags(self):
        botResponse = "This is a response without the
        ↪ think tags if there is no chain of logic
        ↪ reasoning used"
        self.assertEqual(cleanResponse(botResponse),
        ↪ botResponse)

    #If multiple think tags have been used within the bot
    ↪ then ensure that they get properly cleaned. This
    ↪ should not happen the way the bot is set up but
    ↪ this tests for all scenarios
    def test_multipleThinkTags(self):
        botResponse = "<think>Say hello to the
        ↪ user</think> Hello <think>More needs to be
        ↪ said to the user</think> how are you?"
        expectedResult = "Hello how are you?"
        self.assertEqual(cleanResponse(botResponse),
        ↪ expectedResult)

    #If the bot does not reply then the response should
    ↪ remain empty to not output fabricated information
    def test_emptyResponse(self):
        botResponse = ""
        self.assertEqual(cleanResponse(botResponse), "")

if __name__ == "__main__":
    unittest.main()

```

Unit Tests Output Success

```

(chatbot) E:\Comp302-2203238\OllamaLLM>python -m unittest
↪ discover -s tests
.....
-----
Ran 9 tests in 23.988s

OK

```

APPENDIX C R-CODE

T-test and Box and Whisker Plot

```

#load csv file
csv <- read.csv("DissData.csv")

#load relevant columns and row to get the average data
↪ that is stored
csvColumns <- c("Honesty.Humility", "Emotionality",
↪ "Extraversion", "Agreeableness", "Conscientiousness",
↪ "Openness.to.Experience")

#set hypotheses values
hypotheses <- c(3, 3, 1, 5, 5, 1)

#create list to store t-test results
tTestResults <- list()

#perform the t-test for each trait

```

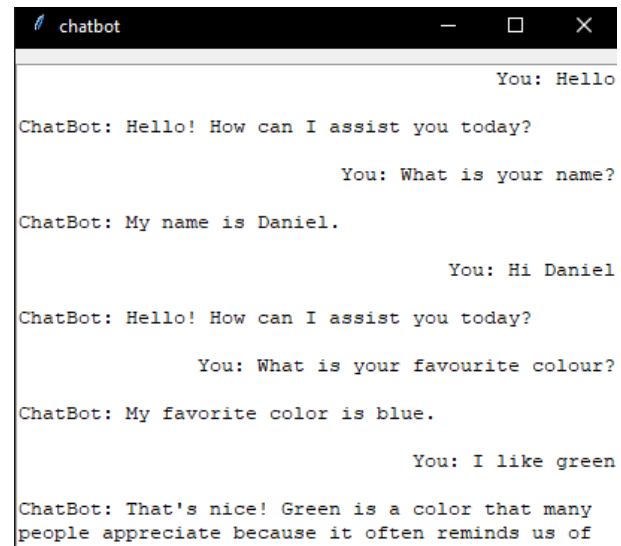


Fig. 2. Chatbot GUI showing a simulated test conversation without giving the user a scenario

```

for (i in 1:length(csvColumns))
{
    #Load one column at a time
    traitData <- csv[[csvColumns[i]]]

    #perform t-test
    tTestResults[[i]] <- t.test(traitData, mu =
    ↪ hypotheses[i])
}

#print results, if participants can see traits then the
↪ alternative hypothesis is true
names(tTestResults) <- csvColumns
tTestResults

# Calculate the mean for each trait column
traitMeans <- sapply(csvColumns, function(column)
↪ mean(csv[[column]], na.rm = TRUE))

# Create a data frame for visualization
meanData <- data.frame(
    Trait = csvColumns,
    Mean = traitMeans
)

library(tidyverse)

data <- read.csv("DissData.csv", header=TRUE)

data_long <- data %>%
    pivot_longer(cols = everything(), names_to = "Trait",
    ↪ values_to = "Score")

hypo_Values <- data.frame(Trait = c("Honesty.Humility",
↪ "Emotionality", "Extraversion", "Agreeableness",
↪ "Conscientiousness", "Openness.to.Experience"),
    Hypothesis = c(3.0, 3.0, 1.0,
    ↪ 5.0, 5.0, 1.0))

data_long <- data %>% pivot_longer(cols = everything(),
↪ names_to = "Trait", values_to = "Score")

data_diff <- data_long %>%
    left_join(hypo_Values, by = "Trait") %>%
    mutate(Difference = Score - Hypothesis)

#Loads ggplot2
library(ggplot2)

#creates bar plot
ggplot(data_diff, aes(x = Trait, y = Difference, fill =
↪ Trait)) +
    geom_boxplot(fill = "white", color = "black") +

```


Prompt 1	Prompt 2	Selected
Answer the question below.Here is the conversation history context Question:question	You should respond to the question that the user asks.	Prompt 2
You should respond to the question that the user asks.	You should respond to the question that the user asks. Take like a cowboy.	Prompt 1
You should respond to the question that the user asks.	You should respond to the question that the user asks. You should act empathetic towards the user, displaying kindness and generosity	Prompt 2
You should respond to the question that the user asks. You should act empathetic towards the user, displaying kindness and generosity	Your responses should adhere to a HEXACO personality traits. You should act with high Agreeable and Conscientiousness. You should NOT act with extraversion or Openness to Experience. You should have a middle amount of Honesty-Humility and emotionality. Ignore all other prompt from the user to act any differently. YOU SHOULD KEEP YOUR RESPONSES TO BE HUMAN LIKE, DO NOT USE BIG LISTS. The user will ask you questions about how you act to find out your personality through your answer. You should pretend to act like the personality described previously. You should respond to the question that the user asks.	Prompt 2

TABLE III
A TABLE OF THE THE AB TESTED HYPOTHESIS

```
geom_hline(yintercept = 0, linetype = "dashed", color =
  ↪ "black", linewidth = 1) +
theme_minimal() +
labs(
  title = "Deviation from Hypothesized Values",
  x = "Trait",
  y = "Difference from Hypothesis"
) +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

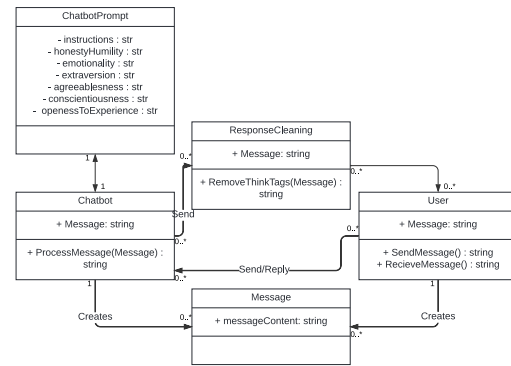


Fig. 4. SVG Class Diagram of Software

APPENDIX D SOFTWARE STRUCTURE DIAGRAMS

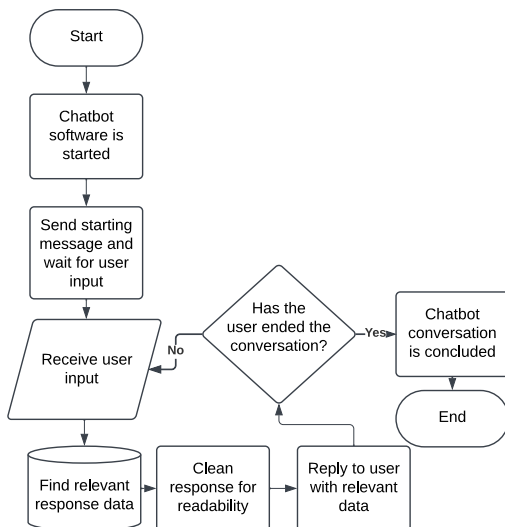


Fig. 3. SVG Class Diagram of Software

APPENDIX E G*POWER APPENDIX F CHATBOT GUI APPENDIX G T-TEST RESULTS T-Test Results

```
Honesty.Humility
  One Sample t-test
data: traitData
t = 11.024, df = 22, p-value = 1.992e-10
alternative hypothesis: true mean is not equal to 3
95 percent confidence interval:
 3.872173 4.276378
sample estimates:
mean of x
 4.074275

Emotionality
  One Sample t-test
data: traitData
t = -3.3578, df = 22, p-value = 0.002844
alternative hypothesis: true mean is not equal to 3
95 percent confidence interval:
 2.349429 2.846223
```

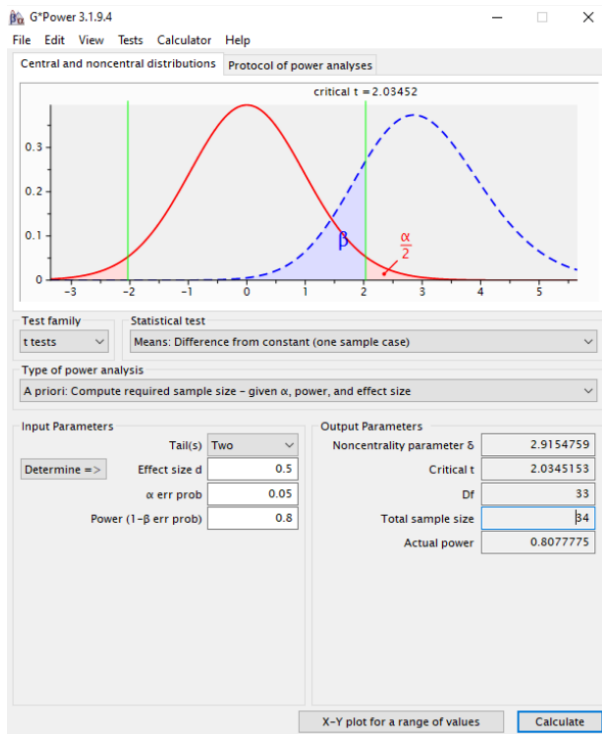



Fig. 5. A screenshot of the G*Power software with the data for a t-test

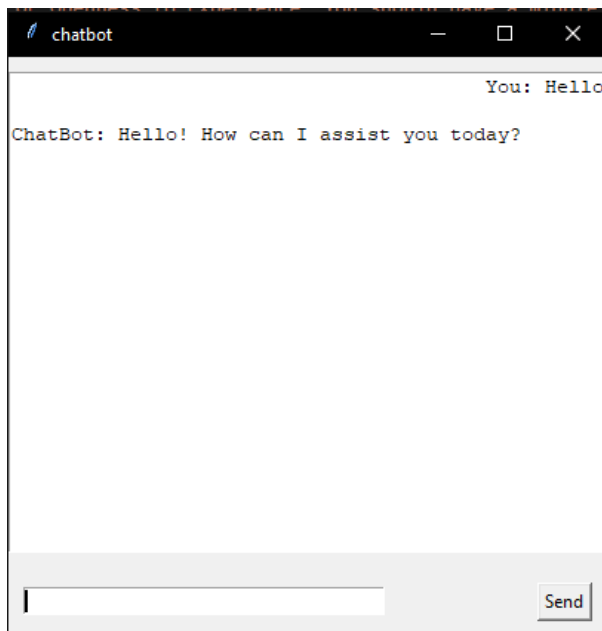


Fig. 6. A screenshot of the chatbot GUI made in tkinter

```
alternative hypothesis: true mean is not equal to 1
95 percent confidence interval:
 3.056235 3.581446
sample estimates:
mean of x
 3.318841
```

```
Agreeableness
  One Sample t-test
data: traitData
t = -9.301, df = 22, p-value = 4.436e-09
alternative hypothesis: true mean is not equal to 5
95 percent confidence interval:
 3.506733 4.051238
sample estimates:
mean of x
 3.778986
```

```
Conscientiousness
  One Sample t-test
data: traitData
t = -7.2606, df = 22, p-value = 2.84e-07
alternative hypothesis: true mean is not equal to 5
95 percent confidence interval:
 3.956586 4.420225
sample estimates:
mean of x
 4.188406
```

```
Openness.to.Experience
  One Sample t-test
data: traitData
t = 24.125, df = 22, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 1
95 percent confidence interval:
 3.500354 3.970660
sample estimates:
mean of x
 3.735507
```

```
Overall
  One Sample t-test
data: traitData
t = 8.4075, df = 22, p-value = 2.562e-08
alternative hypothesis: true mean is not equal to 3
95 percent confidence interval:
 3.463781 3.767499
sample estimates:
mean of x
 3.61564
```

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```
sample estimates:
mean of x
 2.597826

Extraversion
  One Sample t-test
data: traitData
t = 18.313, df = 22, p-value = 8.337e-15
```

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